

Problem-Solution Fit

Date	15 March 2026
Team ID	SWTID-2026-4683
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of the target learners.

1. CUSTOMER SEGMENT(S) [CS] School and college students, self-learners, and learners who need quick help understanding academic topics.	6. CUSTOMER LIMITATIONS Limited study time; difficulty finding simple explanations; varying prior knowledge; dependence on scattered learning resources; limited access to personalized guidance.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Search engines: broad information but not always concise or personalized. Video tutorials: useful demonstrations but time-consuming to find the right level. General AI assistants: flexible, but may not provide a focused learning workflow.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty understanding complex concepts. • Long study material is difficult to revise. • Creating practice questions takes time. • Learners may not know what to study next. Frequency: recurring during study and revision.	9. PROBLEM ROOT / CAUSE [RC] Learning resources are often fragmented and not adapted to the learner's immediate question, knowledge level, or study goal. Manual summarization, quiz creation and planning also consume time.	7. BEHAVIOR + ITS INTENSITY [BE] Learners search, read, ask questions, watch explanations, summarize notes and practice with questions. Intensity: high during assignments, exams and revision.
3. TRIGGERS TO ACT [TR] • Confusing topic or concept. • Upcoming test/assignment. • Need for quick revision. • Need for practice questions or a study plan.	10. YOUR SOLUTION [SL] EduGenie is a Google Gemini powered learning assistant that provides Q&A, simplified concept explanations, quizzes, summaries and personalized learning recommendations through a simple web interface.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Web browser / EduGenie interface FastAPI backend Gemini API OFFLINE / LOCAL Local development and testing with Uvicorn and the lightweight LaMini-Flan-T5 model.
4. EMOTIONS BEFORE / AFTER [EM] Before: confused, time-pressed, uncertain about the right resource. After: clearer understanding, quicker revision, and more structured learning support.	.	.

Fit Summary: EduGenie combines multiple common study-support tasks in one learning workflow, reducing the need to switch between separate tools for explanation, Q&A, revision, practice and learning-path guidance.